Exercise 15C

Solution 1: Radius of the base, $r = 5.25$ cm

Slant height, $l = 10$ cm

$\therefore$ Curved surface area of the cone $= \pi rl$

$$= \frac{22}{7} \times 5.25 \times 10$$

$$= \frac{22}{7} \times \frac{525}{100} \times 10$$

$$= 22 \times \frac{75}{10}$$

$$= \frac{1650}{10}$$

$$= 165 \text{ cm}^2$$

Thus, the curved surface area of the cone is 165 cm^2.

Solution 2: Slant height, $l = 21$ m

Radius of the base, $r = 24/2 = 12$ m

$\therefore$ Total surface area of the cone

$= \pi r(r + l)$

$= \frac{22}{7} \times 12 \times (12 + 21)$

$= \frac{22}{7} \times 12 \times 33$

$= \frac{8712}{7}\ cm^2$

$= 1244.57\ cm^2$

Thus, the total surface area of the cone is 1244.57 cm^2.

Solution 3: Radius of the cone, $r = 7$ cm

Height of the cone, $h = 24$ cm

$\therefore$ Slant height of the cone,

$l = \sqrt{r^2 + h^2}$

$\Rightarrow l = \sqrt{7^2 + (24)^2}$

$\Rightarrow l = \sqrt{49 + 576}$

$\Rightarrow l = \sqrt{625}$

$\Rightarrow l = 25$

Area of the sheet required to make one cap

= Curved surface area of the cone

$= \pi rl$

$= \frac{22}{7} \times 7 \times 25$

$= 22 \times 25$

$= 550 \text{ cm}^2$

$\therefore$ Area of the sheet required to make 10 such caps $= 550 \times 10 = 5500 \text{ cm}^2$

Thus, the area of the sheet required to make 10 such jocker caps is 5500 cm^2.

Solution 4: Slant height, $l = 14$ cm

Let the radius of the base be r cm.

Curved surface area of the cone $= 308 \text{ cm}^2$ (Given)

$\therefore \pi rl = 308$

$\Rightarrow \frac{22}{7} \times r \times 14 = 308$

$\Rightarrow r = \frac{308 \times 7}{22 \times 14}$

$\Rightarrow r = \frac{14 \times 7}{14}$

$\Rightarrow r = 7\ cm$

$\therefore$ Total surface area of the cone $= \pi r(r+l)$

$$= \frac{22}{7} \times 7 \times (7 + 14)$$

$$= 22 \times 21$$

$$= 462 \text{ cm}^2$$

Thus, the radius of the base is 7 cm and the total surface area of the cone is 462 cm^2.

Solution 5: Slant height of the conical tomb, $l = 25$ m

Radius of the conical tomb, $r = \frac{14}{2} = 7$ m

$\therefore$ Curved surface area of the conical tomb $= \pi rl$

$$= \frac{22}{7} \times 7 \times 25$$

$$= 22 \times 25$$

$$= 550 \text{ m}^2$$

Rate of whitewashing $= ₹ 12$ per m^2

$\therefore$ Cost of whitewashing the conical tomb

$=$ Curved surface area of the conical tomb $\times$ Rate of whitewashing

$= 550 \times 12$

$= ₹ 6600$

Thus, the cost of whitewashing the conical tomb is ₹ 6600.

Solution 6: Radius of the conical tent, $r = 24$ m

Height of the conical tent, $h = 10$ m

$\therefore$ Slant height of the conical tent,

$l = \sqrt{r^2 + h^2}$

$\Rightarrow l = \sqrt{(24)^2 + (10)^2}$

$\Rightarrow l = \sqrt{576 + 100}$

$\Rightarrow l = \sqrt{676}$

$\Rightarrow l = 26$ m

$\therefore$ Curved surface area of the conical tent $= \pi rl$

$$= \frac{22}{7} \times 24 \times 26$$

The cost of 1 m^2 canvas is ₹ 70.

$\therefore$ Cost of canvas required to make the tent

$=$ Curved surface area of the conical tent $\times$ ₹ 70

$= \frac{22}{7} \times 24 \times 26 \times 70$

$= 22 \times 24 \times 26 \times 10$

$=$ ₹ 1,37,280

Thus, the cost of canvas required to make the tent is ₹ 1,37,280.

Solution 7:

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Radius of each cone, $r = \frac{}{2} = 20$ cm $= 0.2$ m

Height of each cone, $h = 1$ m

$\therefore$ Slant height of each cone,

$l = \sqrt{r^2 + h^2}$

$\Rightarrow l = \sqrt{(0.2)^2 + 1^2}$

$\Rightarrow l = \sqrt{0.04 + 1}$

$\Rightarrow l = \sqrt{1.04}$

$\Rightarrow l = 1.02$ m

Curved surface area of each cone $= \pi rl$

$= 3.14 \times 0.2 \times 1.02$

$= 0.64056$ m^2

$\therefore$ Curved surface area of 50 cones $= 0.64056 \times 50 = 32.028$ m^2

Cost of painting = ₹ 25 per m^2

∴ Total cost of painting all the cones

= Curved surface area of 50 cones × ₹ 25

= 32.028 × 25

= ₹ 800.70

Thus, the cost of painting all the cones is ₹ 800.70.

Solution 8: Radius of the cone, $r = 35$ cm

Height of the cone, $h = 12$ cm

∴ Slant height of the cone,

$l = \sqrt{r^2 + h^2}$

$\Rightarrow l = \sqrt{(35)^2 + (12)^2}$

$\Rightarrow l = \sqrt{1225 + 144}$

$\Rightarrow l = \sqrt{1369}$

$\Rightarrow l = 37$ cm

(i) Volume of the cone $= \frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \frac{22}{7} \times (35)^2 \times 12$

$= 15400\ cm^3$

(ii) Curved surface area of the cone $= \pi rl$

$= \frac{22}{7} \times 35 \times 37$

$= 4070\ cm^2$

(iii) Total surface area of the cone $= \pi r(r + l)$

$= \frac{22}{7} \times 35 \times (35 + 37)$

$= 22 \times 5 \times 72$

$= 7920 \text{ cm}^2$

Solution 9: Height of the cone, $h = 6$ cm

Slant height of the cone, $l = 10$ cm

Radius,

$r = \sqrt{l^2 - h^2}$

$\Rightarrow r = \sqrt{(10)^2 - (6)^2}$

$\Rightarrow r = \sqrt{100 - 36}$

$\Rightarrow r = \sqrt{64}$

$\Rightarrow r = 8$ cm

Volume of the cone $= \frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times 3.14 \times (8)^2 \times 6$

$= 3.14 \times 64 \times 2$

$= 401.92 \text{ cm}^3$

Curved surface area of the cone $= \pi rl$

$= 3.14 \times 8 \times 10$

$= 251.2 \text{ cm}^2$

$\therefore$ Total surface area $= \pi r\,(l+r)$

$= \pi r(10+8)$

$= (3.14 \times 8 \times 18) \text{ cm}^2$

$= 452.16 \text{ cm}^2$

Solution 10:

Radius of the conical pit, $r = \frac{3.5}{2}$

Depth of the conical pit, $h = 12$ m

$\therefore$ Capacity of the conical pit $= \frac{1}{3}\pi r^2 h$

$$= \frac{1}{3} \times \frac{22}{7} \times \left(\frac{3.5}{2}\right)^2 \times 12$$

$$= \frac{22}{7} \times \left(\frac{35}{2 \times 10}\right)^2 \times 4$$

$$= \frac{22}{7} \times \left(\frac{35}{20}\right)^2 \times 4$$

$$= 22 \times \frac{5}{20} \times \frac{35}{5}$$

$$= 11 \times \frac{1}{10} \times 35$$

$$= \frac{385}{10}$$

$$= 38.5 \text{ m}^3$$

$$= 38.5 \text{ kL} \qquad (\because 1 \text{ m}^3 = 1 \text{ kilolitre})$$

Thus, the capacity of the conical pit is 38.5 kL.

Solution 11: Radius of the heap, $r = 9/2$ m $= 4.5$ m

Height of the heap, $h = 3.5$ m

$\therefore$ Volume of the heap of wheat $= \frac{1}{3}\pi r^2 h$

$$= \frac{1}{3} \times 3.14 \times (4.5)^2 \times 3.5$$

$$= 74.1825 \text{ m}^3$$

Now,

Slant height of the heap,

$l = \sqrt{r^2 + h^2}$

$\Rightarrow l = \sqrt{(4.5)^2 + (3.5)^2}$

$\Rightarrow l = \sqrt{20.25 + 12.25}$

$\Rightarrow l = \sqrt{32.5}$

$\Rightarrow l \approx 5.7\ m$

$\therefore$ Area of the canvas cloth required to just cover the heap of wheat

$=$ Curved surface area of the heap of wheat

$= \pi rl$

$= 3.14 \times 4.5 \times 5.7$

$= 80.541\ m^2$

Thus, the area of canvas cloth required to just cover the heap is approximately 80.541 m^2.

Solution 12: Area of the canvas $= 551\ m^2$

Area of the canvas used in stitching margins and wastage incurred while cutting $= 1\ m^2$

$\therefore$ Area of the canvas used in making the tent $= 551 - 1 = 550\ m^2$

Radius of the tent, $r = 7$ m

Let the slant height and height of the tent be l m and h m, respectively.

Now,

Area of the canvas used in making the tent $= 550\ m^2$

$\therefore \pi rl = 550$

$\Rightarrow \frac{22}{7} \times 7 \times l = 550$

$\Rightarrow l = \frac{550}{22}$

$\Rightarrow l = 25\ m$

Now,

Height of the tent,

$h = \sqrt{l^2 - r^2}$

$\Rightarrow h = \sqrt{(25)^2 - (7)^2}$

$\Rightarrow h = \sqrt{625 - 49}$

$\Rightarrow h = \sqrt{576}$

$\Rightarrow h = 24\ \text{m}$

$\therefore$ Volume of the tent $= \frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \frac{22}{7} \times (7)^2 \times 24$

$= 22 \times 7 \times 8$

$= 1232\ \text{m}^3$

Thus, the volume of the tent that can be made with the given canvas is $1232\ \text{m}^3$.

Solution 13: Radius of the conical tent, $r = 7$ m

Height of the conical tent, $h = 24$ m

Now,

$l = \sqrt{r^2 + h^2}$

$\Rightarrow l = \sqrt{(7)^2 + (24)^2}$

$\Rightarrow l = \sqrt{49 + 576}$

$\Rightarrow l = \sqrt{625}$

$\Rightarrow l = 25\ m$

Curved surface area of the cone $= \pi r l$

$= \frac{22}{7} \times 7 \times 25$

$= 22 \times 25$

$= 550\ \text{m}^2$

Here, area of the cloth =curved surface area of the cone $= 550\ \text{m}^2$

Width of the cloth $= 2.5$ m

$\therefore$ Length of the cloth

$$= \frac{\text{area of the cloth}}{\text{width of the cloth}}$$

$$= \frac{550}{2.5}$$

$$= 220 \text{ m}$$

Solution 14: Let the heights of the first and second cones be h and $3h$, respectively .

Also, let the radius of the first and second cones be $3r$ and r, respectively.

$$\therefore \text{ Ratio of volumes of the cones} = \frac{\text{volume of the first cone}}{\text{volume of the second cone}}$$

$$= \frac{\frac{1}{3}\pi \times (3r)^2 \times h}{\frac{1}{3}\pi \times r^2 \times 3h}$$

$$= \frac{9r^2h}{3r^2h}$$

$$= 3 : 1$$

Solution 15: Suppose that the respective radii and height of the cone and the cylinder are r and h.

Then ratio of curved surface areas $= 8 : 5$

Let the curved surfaces areas be $8x$ and $5x$.

$\therefore 2\pi rh = 8x$

$\Rightarrow 4\pi^2r^2h^2 = 64x^2$ ... (i) [squaring both sides]

and

$\pi rl = 5x$

$\Rightarrow \pi r\sqrt{h^2} + r^2 = 5x$

$\Rightarrow \pi^2r^2(h^2 + r^2) = 25x^2$ ...(ii) [squaring both sides]

Now, dividing equation (i) by (ii), we get

$$\frac{4\pi^2r^2h^2}{\pi^2r^2(h^2 + r^2)} = \frac{64x^2}{25x^2}$$

$$\Rightarrow \frac{4h^2}{(h^2+r^2)} = \frac{64}{25}$$

$$\Rightarrow \frac{h^2}{(h^2+r^2)} = \frac{64}{25\times 4}$$

$$\Rightarrow \frac{h^2}{(h^2+r^2)} = \frac{16}{25}$$

$$\Rightarrow 25h^2 = 16h^2 + 16r^2$$

$$\Rightarrow 16r^2 = 9h^2$$

$$\Rightarrow \frac{r^2}{h^2} = \frac{9}{16}$$

$$\Rightarrow \left(\frac{r}{h}\right)^2 = \left(\frac{3}{4}\right)^2$$

$$\Rightarrow \frac{r}{h} = \frac{3}{4}$$

∴ The ratio of the radius and height of the cone is 3 : 4.

Solution 16: Let the cone which is being melted be denoted by cone 1 and let the cone into which cone 1 is being melted be denoted by cone 2.

Height of cone 1 = 3.6 cm

Radius of the base of cone 1 = 1.6 cm

Radius of the base of cone 2 = 1.2 cm

Let h cm be the height of cone 2.

The volumes of both the cones should be equal.

$$i.e., \frac{1}{3}\pi \times 1.6^2 \times 3.6 = \frac{1}{3}\pi \times 1.2^2 \times h$$

$$\Rightarrow h = \frac{1.6 \times 1.6 \times 3.6}{1.2 \times 1.2}$$

$$\Rightarrow h = 6.4\ cm$$

∴ Height of cone 2 = 6.4 cm

Solution 17: Radius of the tent, r = 52.5 m

Height of the cylindrical portion of the tent, H = 3 m

Slant height of the conical portion of the tent, $l = 53$ m

The tent is a combination of a cylindrical and a conical portion.

Now,

Area of the canvas

= curved surface area of the cone + curved surface area of the cylinder

$= \pi rl + 2\pi rH$

$= \pi r(l + 2H)$

$= \frac{22}{7} \times \frac{105}{2}(53 + 2 \times 3)$

$= \frac{22}{7} \times \frac{105}{2} \times (53 + 6)$

$= 11 \times 15 \times 59$

$= 9735\ m^2$

But area of the canvas = length × breadth

$\therefore$ Length of the canvas $= \frac{\text{area}}{\text{breadth}} = \frac{9735}{5} = 1947$ m

Solution 18: Height of the cylindrical portion of the iron pillar, $h = 2.8$ m $= 280$ cm

Radius of the cylindrical portion of the iron pillar, $r = 20$ cm

Height of the cone which is surmounted on the cylindrical portion, $H = 42$ cm

Now, volume of the pillar

= volume of the cylindrical portion + volume of the conical portion

$= \pi r^2 h + \frac{1}{3}\pi r^2 H$

$= \pi r^2 \left(h + \frac{1}{3}H\right)$

$= \frac{22}{7} \times (10)^2(280 + \frac{1}{3} \times 42)$

$= \frac{22}{7} \times 100(280 + 14)$

$= \frac{22}{7} \times 100 \times 294$

$= 22 \times 100 \times 42$

$= 92400 \text{ cm}^3$

$\therefore$ Weight of the pillar = volume of the pillar $\times$ weight per cubic cm

$$= 92400 \times \frac{7.5}{1000}$$

$$= 924 \times \frac{7.5}{10}$$

$$= \frac{6930}{10}$$

$$= 693 \text{ kg}$$

Solution 19: Height of the cylinder $= 10$ cm

Radius of the cylinder $= 6$ cm

The respective heights and radii of the cone and the cylinder are the same.

$\therefore$ Volume of the remaining solid = volume of the cylinder $-$ volume of the cone

$$= \pi r^2 h - \frac{1}{3}\pi r^2 h$$

$$= \frac{2}{3}\pi r^2 h$$

$$= \frac{2}{3} \times 3.14 \times 6^2 \times 10$$

$$= \frac{2}{3} \times 3.14 \times 36 \times 10$$

$$= 2 \times 3.14 \times 12 \times 10$$

$$= 753.6 \text{ cm}^3$$

Solution 20: Diameter of the cylindrical pipe $= 5$ mm

Radius of the cylindrical pipe $= 5/2 = 2.5$ mm $= 0.25$ cm

Water flowing per minute $= 10$ m $= 1000$ cm

Volume of water flowing per minute through the cylindrical pipe

$= \pi \times (0.25)^2 \times 1000 \text{ cm}^3$

Radius of the conical vessel $= 40/2 = 20$ cm

Depth of the vessel $= 24$ cm

Volume of the conical vessel $= \frac{1}{3}\pi \times (40)^2 \times 24 \text{ cm}^3$

Let the time taken to fill the conical vessel be x min.

Volume of water flowing per minute through the cylindrical pipe $\times x =$ volume of the conical vessel

$$\Rightarrow x = \frac{\text{volume of the conical vessel}}{\text{Volume of water flowing per minute through the cylindrical pipe}}$$

$$\Rightarrow x = \frac{\frac{1}{3}\pi \times (20)^2 \times 24}{\pi \times (0.25)^2 \times 1000}$$

$$\Rightarrow x = \frac{400 \times 24}{3 \times 0.0625 \times 1000}$$

$\Rightarrow x = 51.2\ min$

$\Rightarrow x = 51\ min\ 12\ sec$

∴ The cylindrical pipe would take 51 min 12 sec to fill the conical vessel.

Solution 21: Radius of the conical tent, $r = 5$ m

Area of the base of the conical tent $= \pi r^2 = \frac{22}{7} \times 5^2 = \frac{550}{7}$

Average area occupied by a student on the ground $= \frac{5}{7}\ \text{m}^2$

∴ Number of students who can sit in the tent

$$= \frac{\text{Area of the base of the conical tent}}{\text{Average area occupied by a student on the ground}}$$

$$= \frac{\frac{550}{7}}{\frac{5}{7}}$$

$= \frac{550}{5}$

$= 110$

Thus, the number of students who can sit in the tent is 110.

Let the slant height of the tent be l m.

Curved surface area of the tent $= 165\ m^2$

$\therefore \pi r l = 165$

$\Rightarrow \frac{22}{7} \times 5 \times l = 165$

$\Rightarrow l = \frac{165 \times 7}{22 \times 5}$

$\Rightarrow l = 10.5\ m$

Let the height of the tent be h m.

$h = \sqrt{l^2 - r^2}$

$\Rightarrow h = \sqrt{(10.5)^2 - 5^2}$

$\Rightarrow h = \sqrt{110.25 - 25}$

$\Rightarrow h = \sqrt{85.25}$

$\Rightarrow h \approx 9.23$ m

$\therefore$ Volume of the tent $= \frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \frac{22}{7} \times (5)^2 \times 9.23$

$\approx 241.74\ m^3$

Thus, the volume of the conical tent is approximately $241.74\ m^3$.